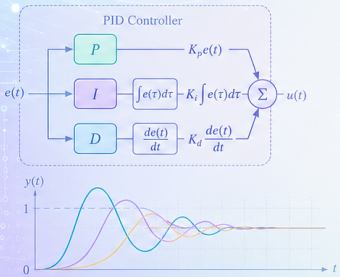
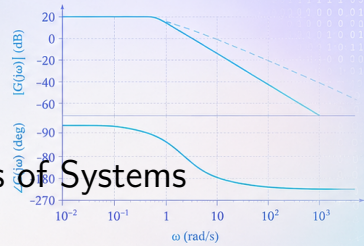


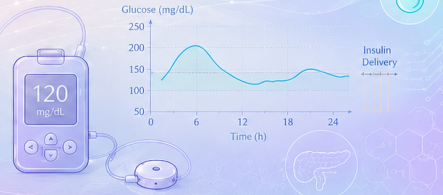
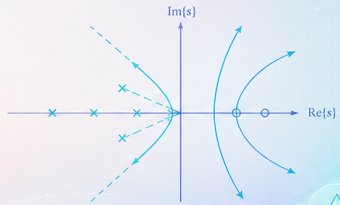
Mathematical Models of Systems

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$$G(s) = \frac{Y(s)}{U(s)} \quad a_n \frac{d^n y(t)}{dt^n} + \dots + a_1 \frac{dy(t)}{dt} + a_0 y(t) = b_m \dots + b_m \frac{d^m u(t)}{dt^m} + \dots + b_1 \frac{du(t)}{dt} + b_0 u(t)$$



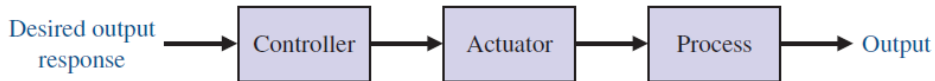
Learning Objectives

- ▶ Mathematical models of physical systems are key elements in the design and analysis of control systems.
- ▶ The dynamic behavior is generally described by ordinary differential equations.
- ▶ Since most physical systems are nonlinear, we consider linearization approximations.
- ▶ We obtain the input–output relationship of the system.

Prerequisites: Basic knowledge of differential equations, Laplace transforms, electrical circuits, and elementary system modelling.

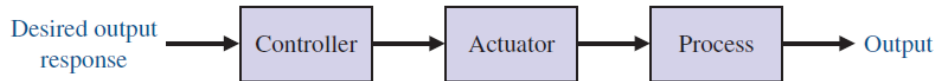
Recap

Open Loop – no attempt to verify the outcome, no feedback, no correction



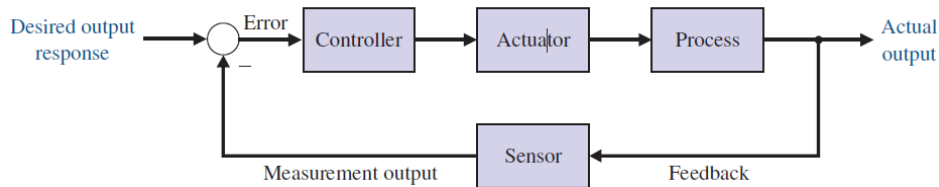
Recap

Open Loop – no attempt to verify the outcome, no feedback, no correction



Closed

Loop – outcome assessed, feedback provided, corrective action taken



Dynamical system

- ▶ A **physical system** represents a physical (electrical or biological or chemical or information, or...) system that we are trying to analyze or design.
- ▶ Physical (**dynamical**) systems usually show dynamic behavior.
- ▶ A system **model** is a mathematical representation of a physical system.
- ▶ Models allow us to make predictions about how a system will behave.
- ▶ There may be multiple models for a single dynamical system.
- ▶ All models are approximations of the real system behavior.
- ▶ A **dynamical system** use one or more mathematical frameworks (e.g., ODEs, PDEs, automata, etc).
- ▶ We will mainly be interested in models of dynamical systems describing the input/output behavior of systems.

Model terminology

In control, the model is given by **differential equation**

$$\frac{d\mathbf{x}}{dt} = \mathbf{f}(\mathbf{x}, \mathbf{u}), \quad \mathbf{y} = \mathbf{h}(\mathbf{x}, \mathbf{u}),$$

where

- ▶ $\mathbf{x}$ is a vector of state variables that captures effects of the past. It consists of physical quantities determine future evolution (absent external excitation).
- ▶ $\mathbf{u}$ is a vector of control signals. These inputs are extrinsic to the system dynamics (externally specified).
- ▶ $\mathbf{y}$ is a vector of measurements These outputs are functions of the state and inputs; they are not independent variables. They are often a subset of the state.

while $\mathbf{f}$ and $\mathbf{h}$ are possibly nonlinear mappings of their arguments to vectors of the appropriate dimensions.

Autonomous System

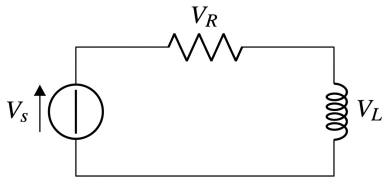
$$\frac{d\mathbf{x}}{dt} = \mathbf{f}(\mathbf{x}, \mathbf{u}), \quad \mathbf{y} = \mathbf{h}(\mathbf{x}, \mathbf{u}),$$

- ▶ The model is called a forced or controlled differential equation.

$$\frac{d\mathbf{x}}{dt} = \mathbf{f}(\mathbf{x}), \quad \mathbf{y} = \mathbf{h}(\mathbf{x}),$$

- ▶ The model is called an autonomous system because there are no external influences

Series RL Circuit: System Model



R : Resistance,

L : Inductance,

$$V_R = Ri,$$

$$V_L = L \frac{di}{dt}.$$

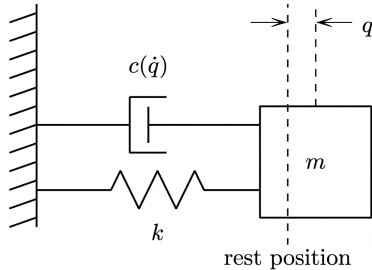
► Kirchhoff's voltage law:

$$V_s - V_R - V_L = 0$$

► System model:

$$L \frac{di}{dt} = V_s - Ri$$

Mass–Spring–Damper System

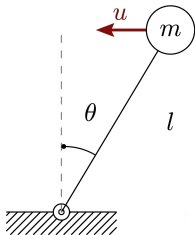


- m : mass,
- F : external force,
- c : friction (damper),
- k : spring stiffness,
- q : deviation from the rest position.

► **System model:** From Newton's second law,

$$m\ddot{q} + c\dot{q} + kq = F. \quad (1)$$

Inverted Pendulum



m : mass,
 l : length,
 u : external force,
 θ : angular displacement.

► **Torque:**

$$T = mgl \sin \theta - ul \cos \theta$$

► **Moment of inertia:**

$$J = ml^2$$

► **System model:** From Newton's second law,

$$ml^2\ddot{\theta} = mgl \sin \theta - ul \cos \theta.$$

State-Space Model

- ▶ The **order of a model** is defined as the dimension of the state vector $\mathbf{x}$.
- ▶ A system is called **time invariant** when the functions f and h do not explicitly depend on time t .
- ▶ In a time-varying system, the functions may explicitly depend on time.
- ▶ The nonlinear state-space model contains two mappings:
 - ▶ f : describes the rate of change of the state variables.
 - ▶ h : describes the measured outputs.

General nonlinear state-space model

$$\dot{\mathbf{x}} = f(\mathbf{x}, \mathbf{u}), \quad \mathbf{y} = h(\mathbf{x}, \mathbf{u}).$$

Linear State-Space Model

- ▶ A model is called a **linear state-space model** when f and h are linear functions of the state $\mathbf{x}$ and the control input $\mathbf{u}$.
- ▶ The standard linear state-space representation is

Linear state-space equations

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{u}, \quad \mathbf{y} = \mathbf{C}\mathbf{x} + \mathbf{D}\mathbf{u}.$$

- ▶ When $\mathbf{A}$, $\mathbf{B}$, $\mathbf{C}$, and $\mathbf{D}$ are constant matrices, the system is called a **linear time-invariant system**, or an **LTI system**.

State-Space Matrices

- ▶ A : **System matrix** or **dynamics matrix**
- ▶ B : **Input matrix** or **control matrix**
- ▶ C : **Output matrix** or **sensor matrix**
- ▶ D : **Direct transmission matrix** or **feedthrough matrix**
- ▶ The matrix D represents the direct effect of the input $\mathbf{u}$ on the output $\mathbf{y}$.
- ▶ In many physical systems,

$$D = 0,$$

which means that the input does not directly influence the output.

Thank You

Questions and Discussion